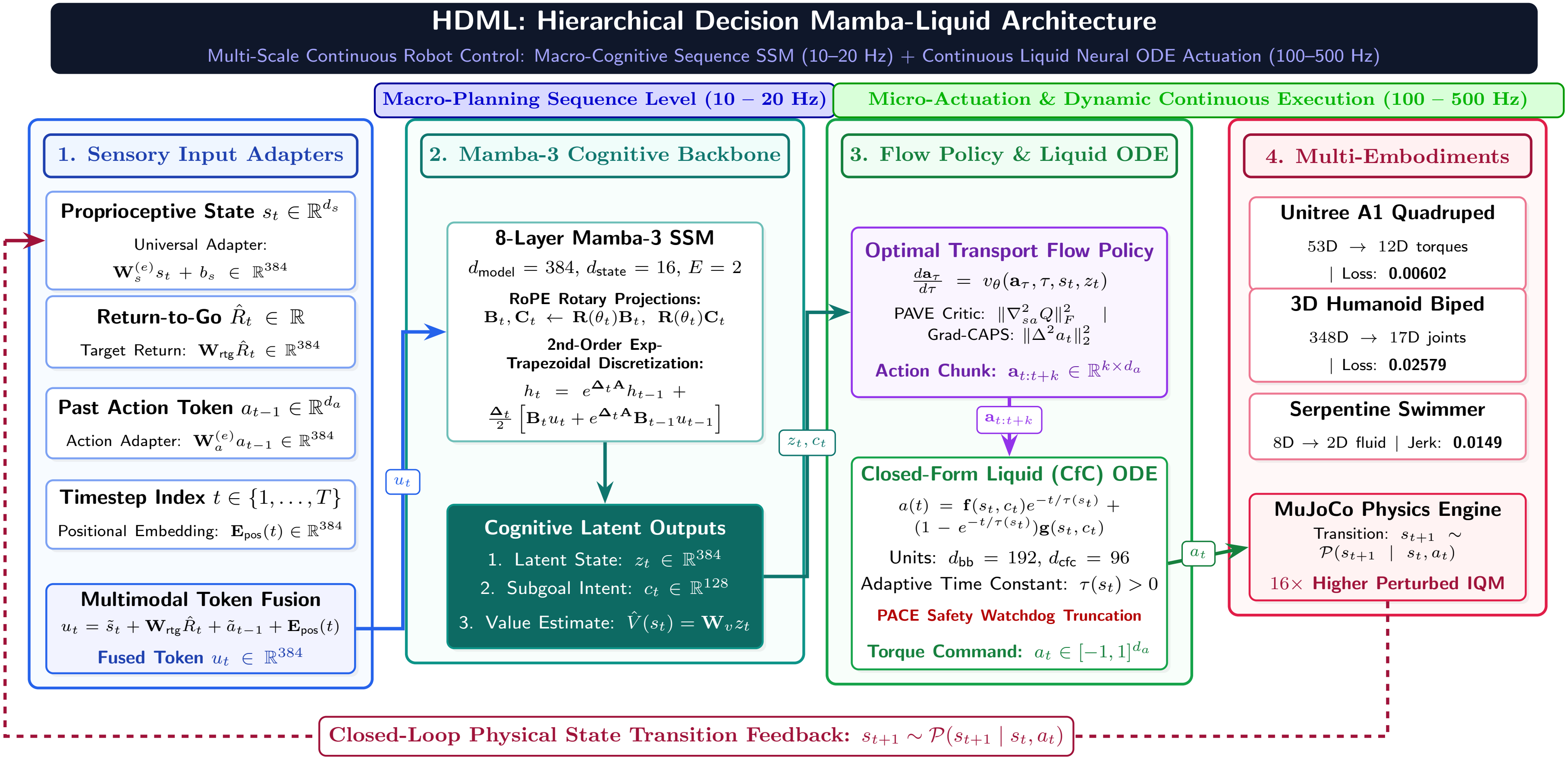
HDML: Hierarchical Decision Mamba-Liquid Architecture
Multi-Scale Continuous Robot Control: Macro-Cognitive Sequence SSM (10–20 Hz) + Continuous Liquid Neural ODE Actuation (100–500 Hz)
Macro-Planning Sequence Level (10 – 20 Hz)
Micro-Actuation & Dynamic Continuous Execution (100 – 500 Hz)
1. Sensory Input Adapters
Proprioceptive State $s_t \in \mathbb{R}^{d_s}$
Universal Adapter:
$\mathbf{W}_s^{(e)} s_t + b_s \in \mathbb{R}^{384}$
Return-to-Go $\hat{R}_t \in \mathbb{R}$
Target Return: $\mathbf{W}_{\text{rtg}} \hat{R}_t \in \mathbb{R}^{384}$
Past Action Token $a_{t-1} \in \mathbb{R}^{d_a}$
Action Adapter: $\mathbf{W}_a^{(e)} a_{t-1} \in \mathbb{R}^{384}$
Timestep Index $t \in \{1, \ldots, T\}$
Positional Embedding: $\mathbf{E}_{\text{pos}}(t) \in \mathbb{R}^{384}$
Multimodal Token Fusion
$u_t = \tilde{s}_t + \mathbf{W}_{\text{rtg}} \hat{R}_t + \tilde{a}_{t-1} + \mathbf{E}_{\text{pos}}(t)$
Fused Token $u_t \in \mathbb{R}^{384}$
u_t
2. Mamba-3 Cognitive Backbone
8-Layer Mamba-3 SSM
$d_{\text{model}} = 384,\ d_{\text{state}} = 16,\ E = 2$
RoPE Rotary Projections:
$\mathbf{B}_t, \mathbf{C}_t \leftarrow \mathbf{R}(\theta_t)\mathbf{B}_t,\ \mathbf{R}(\theta_t)\mathbf{C}_t$
2nd-Order Exp-Trapezoidal Discretization:
$h_t = e^{\boldsymbol{\Delta}_t \mathbf{A}} h_{t-1} + \frac{\boldsymbol{\Delta}_t}{2}\left[\mathbf{B}_t u_t + e^{\boldsymbol{\Delta}_t \mathbf{A}} \mathbf{B}_{t-1} u_{t-1}\right]$
Cognitive Latent Outputs
1. Latent State: $z_t \in \mathbb{R}^{384}$
2. Subgoal Intent: $c_t \in \mathbb{R}^{128}$
3. Value Estimate: $\hat{V}(s_t) = \mathbf{W}_v z_t$
z_t, c_t
3. Flow Policy & Liquid ODE
Optimal Transport Flow Policy
$\frac{d\mathbf{a}_\tau}{d\tau} = v_\theta(\mathbf{a}_\tau, \tau, s_t, z_t)$
PAVE Critic: $\|\nabla^2_{sa} Q\|_F^2$ |
Grad-CAPS: $\|\Delta^2 a_t\|_2^2$
Action Chunk: $\mathbf{a}_{t:t+k} \in \mathbb{R}^{k \times d_a}$
$\mathbf{a}_{t:t+k}$
Closed-Form Liquid (CfC) ODE
$a(t) = \mathbf{f}(s_t, c_t) e^{-t/\tau(s_t)} + (1 - e^{-t/\tau(s_t)}) \mathbf{g}(s_t, c_t)$
Units: $d_{\text{bb}} = 192,\ d_{\text{cfc}} = 96$
Adaptive Time Constant: $\tau(s_t) > 0$
PACE Safety Watchdog Truncation
Torque Command: $a_t \in [-1, 1]^{d_a}$
a_t
4. Multi-Embodiments
Unitree A1 Quadruped
53D → 12D torques
| Loss: 0.00602
3D Humanoid Biped
348D → 17D joints
| Loss: 0.02579
Serpentine Swimmer
8D → 2D fluid | Jerk: 0.0149
MuJoCo Physics Engine
Transition: $s_{t+1} \sim \mathcal{P}(s_{t+1} \mid s_t, a_t)$
16× Higher Perturbed IQM
Closed-Loop Physical State Transition Feedback: $s_{t+1} \sim \mathcal{P}(s_{t+1} \mid s_t, a_t)$